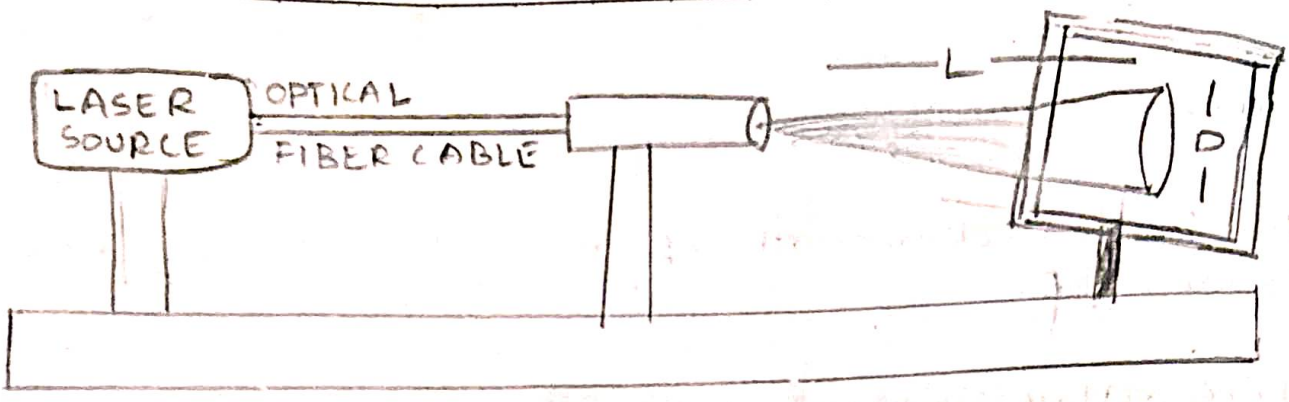
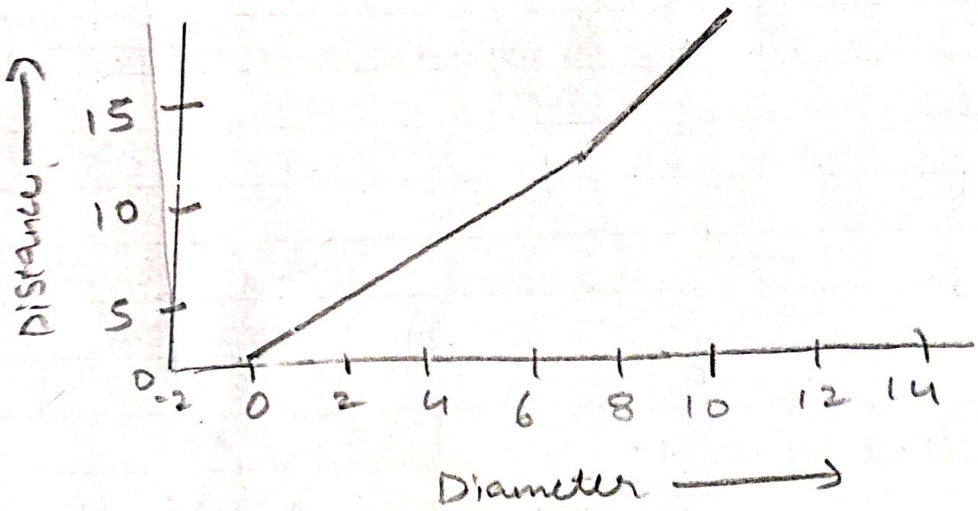


DATATABLE

Sr. No	Dist. of screen (L)	Diameter (D)
1	0	0
2	4	4.444
3	8	8.888
4	12	13.332



* Aim:- To determine the ~~na~~ numerical Aperture (NA) of an optical fibre.

* Apparatus:- optical fiber, laser source/ LED source, screen, Fiber holder, scale/measuring tape, power supply.

* Theory:- opticals fibers transmit light based on the principle of Total internal reflection. Light entering the fiber within a certain angle is guided through the ~~one~~ core.

The NA indicates the light-gathering ability of the optical fiber.
It is defined as:-

$$\boxed{NA = \sin\theta}$$

where,

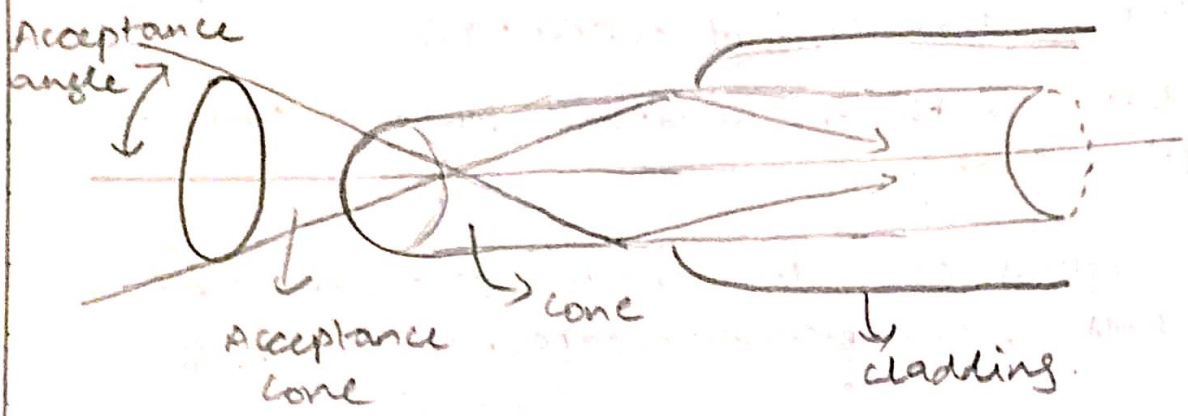
θ = Acceptance angle of fiber.

Experimentally,

$$\boxed{NA = \frac{D}{\sqrt{4L^2 + D^2}}}$$

→ Larger NA means the fiber can accept light from a wider angle.

* Procedure:-
1. Connect the optical fiber to the light source.
2. Place the other end of the fiber facing a screen.
3. Measure the distance (L) between the fiber end and the screen.


Acceptance angle
Acceptance cone
core
cladding

4. Switch on the Laser / LED source so light passes through the fiber.
5. A circular light spot will appear on the screen.
6. Measure the diameter (D) of the spot using a scale.
7. Repeat the readings for accuracy.
8. Calculate the Numerical Aperture using the formula.

* Observation :-
(In white page)

* Result :-

$$\text{Slope} = \frac{(y_2 - y_1)}{(x_2 - x_1)} = \frac{8-4}{8.888 - 4.444} = \frac{4}{4.444} = 0.90$$

$L/D = 0.90$

$D/L = \frac{1}{0.90} = 1.111$

so, $\frac{D}{2L} = 0.555$

tan (Inverse) $0.555 = 0.507$ = Acceptance Angle.

$\sin 0.507 = 0.485$ = Numerical Aperture.

$\boxed{NA = 0.485}$